

Employee Turnover Analytics

Predicting and Preventing Employee Attrition — A Plain-English Overview

Portobello Tech — HR Department

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Summary

We built a tool that looks at an employee's day-to-day work patterns — satisfaction, performance ratings, workload, tenure — and predicts how likely they are to leave the company. It also groups the employees who already left into a few distinct “why they left” stories, so HR can respond with the right kind of support instead of a one-size-fits-all approach.

92%

of employees who actually leave are
correctly flagged in advance

3

distinct “why they leave” employee profiles
identified

20%

of the raw data was duplicate records —
found and removed

What Problem This Solves

- **The cost of surprise:** Today, HR typically learns someone is leaving only when they hand in their resignation — too late to do anything about it.
- **Losing good people is expensive:** Recruiting, hiring, and training a replacement costs far more than most retention efforts would have cost.
- **This project shifts HR from reactive to proactive:** instead of reacting after someone quits, we can flag risk before a resignation lands and act while there's still a chance to change the outcome.

What Data Was Used

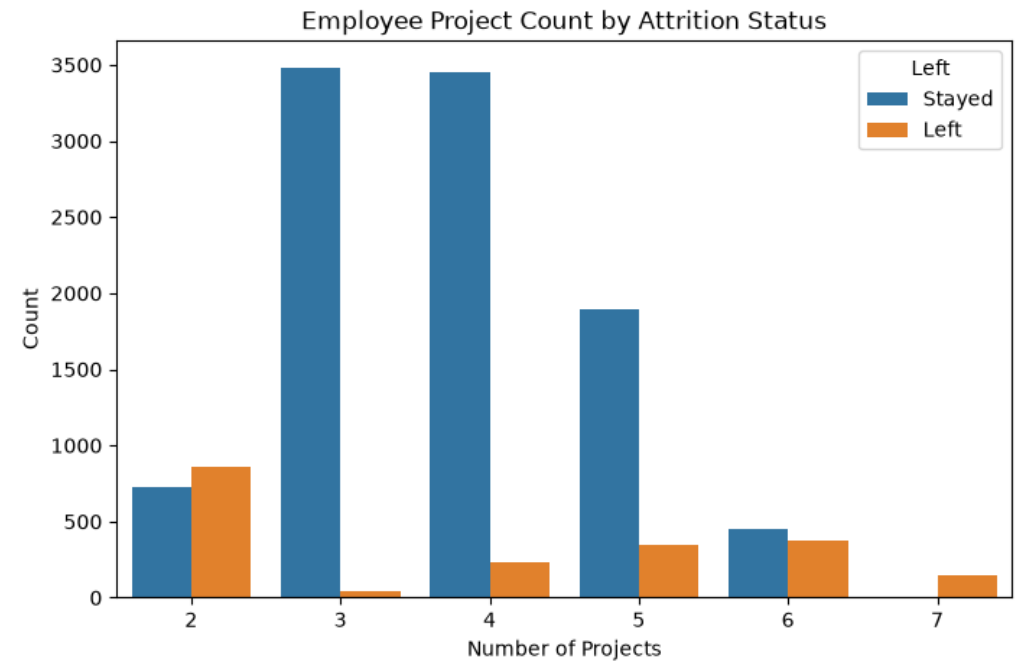
- Approximately 15,000 employee records from Portobello Tech's HR system.
- Inputs used to predict risk: satisfaction survey score, most recent performance evaluation, number of active projects, average hours worked per month, years at the company, department, and salary tier.
- **What the model learns to predict:** historical records of which employees stayed vs. left — this is the outcome the model is trained to forecast, not an input we already know for current employees.
- No sensitive personal information was used — no names, ages, gender, or other personal identifiers were part of this model.

What Was Cleaned Before Modeling

- **Fixed a labeling error:** a column meant to show department was mislabeled “sales” in the raw export, even though it actually contained every department name.
- **Removed duplicate records:** about 3,000 of the 15,000 records (20%) were exact duplicates — left in, they would have unfairly inflated how accurate the model looked.
- **No missing information:** every employee record was complete; nothing had to be discarded for incompleteness.

What the Model Found

- **Satisfaction and tenure matter most:** how happy someone is at work, and how long they've been here, are by far the strongest predictors of who leaves — more than workload or pay alone. (This tells us what best predicts leaving, not a proven cause of it.)
- **Three “why they leave” groups:** (1) burned-out high performers, (2) satisfied people likely leaving for a better opportunity elsewhere, (3) a mixed group we can't yet explain from this data — needs a direct conversation, not a guess.
- **Workload is a U-shape, not a straight line:** employees with too few projects (2) AND employees with too many (6–7) both leave at high rates — 3–4 projects is the sweet spot.



How Well It Worked

92%

Recall — of everyone who actually leaves, this is the share the model correctly flags in advance

86%

Precision — when the model flags someone as at-risk, this is how often it's right

0.98

ROC-AUC — overall ability to tell at-risk employees apart from safe ones (1.0 = perfect, 0.5 = coin flip)

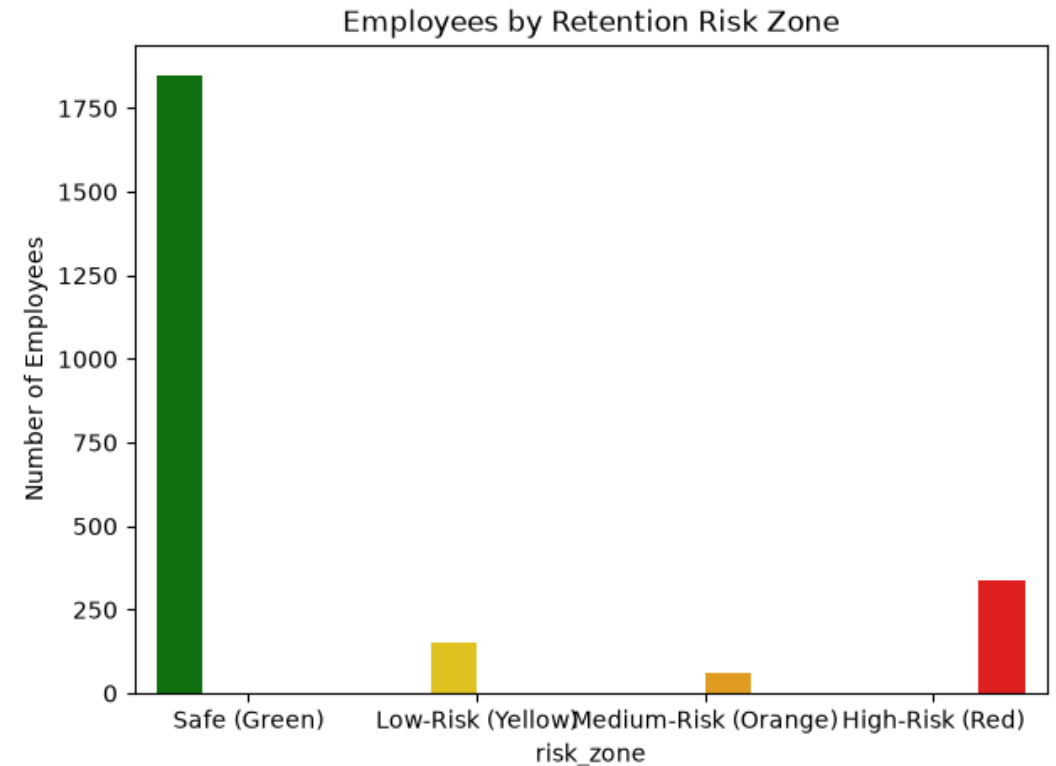
How this was tested: these numbers come from about 2,400 employees the model never saw while it was being built — the honest measure of how it performs on people it hasn't met, not a score on its own homework.

Best use: this tool is best for prioritizing where to focus retention effort across the whole workforce — not for making a final, individual yes/no judgment about any one person.

We chose to prioritize recall (catching actual leavers) over precision (avoiding false alarms), because missing someone who was really about to leave costs far more than occasionally checking in with someone who wasn't.

Turning Scores Into Action: Four Risk Zones

- **Safe (Green):** under 20% predicted risk — no action needed.
- **Low-Risk (Yellow):** 20%–60% predicted risk — light monitoring.
- **Medium-Risk (Orange):** 60%–90% predicted risk — active manager/HR review recommended.
- **High-Risk (Red):** over 90% predicted risk — prioritize immediate outreach.



Every score the model produces gets sorted into one of these four zones — this is how the numbers become action.

What Staff Should Do With Scores

- **Use scores to prioritize, not to judge:** a risk score is a signal to pay closer attention, not a verdict on the employee.
- **High- and Medium-Risk employees:** route to a manager or HR conversation, tailored to the likely “why” group — workload relief for burnout, growth conversations for satisfied high performers, a direct stay-interview for the unclear group.
- **Also consider salary:** turnover drops sharply as pay increases (20% at low pay vs. 5% at high pay) — a compensation review is worth adding for flagged employees on the low salary tier.
- **Safe and Low-Risk employees:** no special action needed — continue business as usual.

What This Should NOT Be Used For

- **Never the sole basis for a negative action:** no denying a raise, no punitive reassignment, no using a score against an employee in any way.
- **Not a performance review substitute:** it predicts turnover risk, not job performance or worth.
- **Not yet validated for fairness:** the data didn't include age, gender, or other protected characteristics, so we can't confirm the tool treats every group equitably — it should not be used in any way that could disproportionately affect a specific group without further review.

Recommended Steps

- Score the active workforce quarterly using the model.
- Route every flagged (Medium/High-Risk) employee through a human manager or HR review before any action is taken.
- Match the response to the likely “why” group rather than applying one generic response to everyone.
- Track whether flagged employees who received support actually stay longer — this proves (or disproves) that the tool is actually helping.

Final Stakeholder Decision

Recommended decision: approve a pilot rollout in a single department for one quarter, with the human-review safeguard in place, before deciding on a company-wide rollout.

One-Page Takeaway

- **The problem:** HR learns who's leaving too late to act. This tool flags risk in advance.
- **The result:** correctly identifies 92% of employees who actually go on to leave, and is right 86% of the time when it raises a flag — measured on employees the model never saw during training.
- **The key drivers:** satisfaction and tenure matter most; workload has a sweet spot (3–4 projects); pay level matters too.
- **The three leaver profiles:** burned-out high performers, satisfied people leaving for growth elsewhere, and an unclear group needing direct conversation.
- **The ask:** approve a one-department pilot, route flagged employees through human review, and measure whether it actually improves retention.